

Hongdao Meng

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EDUCATION

New York University

Master of Science in Computer Science GPA: 3.83

Sep. 2024 - May 2026 (Expected)

New York, NY

Beijing University of Technology

Bachelor of Engineering in Information Security GPA: 4.00

Sep. 2020 - Jul. 2024

Beijing, CN

WORK EXPERIENCE

TikTok

Software Engineer Intern

May 2025 – Sep. 2025

Seattle, WA

- Developed a Python-based **DAG-DSL** migration toolkit (leveraging **Protobuf** and **regex**) to auto-convert legacy configuration files of the **Bytedance Feature Service (BFS)**, a scalable feature extraction service powering TikTok's recommendation system, into standardized DSL definitions, boosting operator migration throughput by 85%.
- Designed and implemented a **Blade**-based **C++** operator migration pipeline for BFS, streamlining the workflow to improve operator migration speed by 83.3%, and personally migrated 20 operators.
- Architected and developed a one-stop workflow for **User Data Accessor**, a data extraction service for recommendation engine, covering service build, auto-generate **Python** operators, DSL compilation, and **RPC** request, enabling collaboration among dozens of engineers and accelerating development by 73.3%.
- Engineered a **C++** debug framework for the DAG-based Service Engine, leveraging **gflags** to control on-demand logging of each node's execution details within topologically ordered DAG subgraphs.

C2SMARTER Center

Software Engineer Intern

Jan. 2025 – Present

New York, NY

- Developed a full-stack **RAG**-based chatbot system using **React**, **Redux**, and **Flask**, delivering responsive UI and seamless communication, improving user engagement by 33.7% and reducing data retrieval latency by 23.6%.
- Built a scalable back-end infrastructure with **Docker**, **MongoDB**, **MilvusDB**, and **RESTful APIs**, enabling efficient data storage and retrieval, which improved query performance by 25.6% and reduced deployment setup time by 15.7%.
- Deployed application on **AWS EC2** and implemented **CI/CD** pipelines with **Jenkins**, ensuring high availability and automated testing, which decreased deployment cycles by 18.7% and supported concurrent users from 5k to 10k.
- Implemented monitoring and observability with **Grafana** and **Prometheus**, providing real-time system insights and reducing mean time to resolution (MTTR) by 25.3%.

DeepFake Detection Startup

Software Engineer, Founder

Sep. 2024 – Dec. 2024

New York, NY

- Developed core modules of a deepfake detection web platform using **React** and **TypeScript** for seamless real-time interaction, enabling 1,200+ concurrent users and reducing client-side rendering latency by 21.3%.
- Built a real-time communication layer using **Django** and **WebSocket** for robust middleware communication, delivering real-time updates with 180ms latency and improving user task efficiency by 25.6%.
- Deployed backend services on **Kubernetes** with **ELB** load balancing, ensuring scalability and reliability.
- Optimized **PostgreSQL** query execution through composite index tuning, reducing average response time by 18%.

QingTeng Cloud Security

Software Engineer Intern

Feb. 2024 – Aug. 2024

Product R&D Department

- Developed a real-time threat detection platform using **Spring Boot** and **MyBatis**, processing 150k+ security logs/day with optimized batch SQL queries, reducing alert latency by **22%** (from 650ms to 500ms P95).
- Designed distributed messaging and caching architecture using **RabbitMQ** and **Redisson**, achieving 120ms latency, 25% higher throughput, and 26% lower database load under peak traffic.
- Implemented comprehensive logging and monitoring system using **ELK Stack (Elasticsearch, Logstash, Kibana)**

SKILLS

Languages: Python, C/C++, Java, Go, SQL, JavaScript, TypeScript, HTML/CSS, Shell, PHP, MATLAB, LaTeX

Frameworks: Blade, React, Vue.js, Django, Flask, Node.js, Spring Boot, PyTorch, TensorFlow, Pandas, MyBatis

DataBase: MySQL, Redis, MongoDB, MilvusDB, PostgreSQL, DynamoDB, Oracle, Firebase, RocketMQ, Elasticsearch

Tools: Git, Docker, AWS, Azure, CMake, Postman, CI/CD, Jenkins, Nginx, LangChain, OpenCV, Jira, Figma, Kibana